

Lecture 10 – Two-species competition

Today: Graphical analysis of two-species competition. Hand out Q3 on pg.2

Next: 2-D stability analysis

Concepts for the road ahead:

Coexistence, Invasibility, Priority effects, Alternative Stable States
Phase diagrams & Zero Net Growth Isoclines (ZNGI's)

Two-species competition

Motivate with Crombie (1946) flour beetle experiments. 2 slides.

Extend 1 sp. logistic to 2 spp.

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1}{K_1} - \alpha_{12} \frac{N_2}{K_1} \right) \quad \frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2}{K_2} - \alpha_{21} \frac{N_1}{K_2} \right)$$

Or in slightly different general form:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i}{K_i} - \alpha_{ij} \frac{N_j}{K_i} \right) \quad \text{for } i \neq j.$$

Contrast *intra*-specific density-dependence (self-limitation) vs. *inter*-specific competition

α_{ij} - effect of 1 average j individual on 1 average i individual relative to effect that i has on self.

= per capita effect

'How much of K_i does each j individual use?'

e.g., if 10 j individuals consume equivalent to 1 i individual, then $\alpha_{ij} = \frac{1}{10}$.

This is a model of **exploitation** with *implicit* resources.

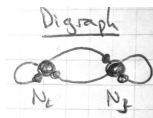
Implicit - Resource dynamics are *not* modeled

- Phenomenological model depiction of resources

- e.g., K - 'carrying capacity'

That is:

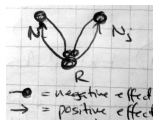
$$\frac{dN_i}{dt} = f_i(N_i, N_j)$$



Explicit - Resource dynamics *are* modeled

e.g. Consumer-resource models (next class)

$$\frac{dN_i}{dt} = f_i(N_i, R) \quad \frac{dR}{dt} = f_R(N_i, N_j, R)$$



Exploitation - Indirect negative interaction through joint use of shared *limiting* resource(s).

Interference - Direct negative interaction preventing use of resource(s).

Today's Q:

When can two species coexist on the same shared resource(s)?

When can Sp. 1 invade a community consisting of Sp. 2?

⇒ How many equilibria are there?

⇒ Are equilibria stable or unstable?

Hand out and work through quiz as group: (a)

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i}{K_i} - \alpha_{ij} \frac{N_j}{K_i} \right)$$

Qualitatively...

$$(N_i^*, N_j^*) = \begin{cases} 0, & 0 \\ N_i^* > 0, & 0 \\ 0, & N_j^* > 0 \\ N_i^* > 0, & N_j^* > 0 \end{cases}$$

(b)

When $N_j = 0 \Rightarrow$ reduces to 1 sp. logistic:

$$\begin{aligned} \frac{dN_i}{dt} &= r_i N_i \left(1 - \frac{N_i}{K_i} \right) \\ N_i^* &= K_i \end{aligned}$$

(c)

Replace i for j , $\Rightarrow N_j^* = K_j$ in absence of i .

(d)

Want $N_i^* > 0$ & $N_j^* > 0$

A: Solve for one as function of the other:

$$\begin{aligned} r_i N_i - \frac{r_i N_i N_i}{K_i} - \frac{r_i \alpha_{ij} N_i N_j}{K_i} &= 0 \\ r_i N_i K_i &= r_i N_i N_i + r_i \alpha_{ij} N_i N_j \quad (\text{move and multiply by } K_i) \\ K_i &= N_i + \alpha_{ij} N_j \quad (\text{divide by } r_i \text{ and } N_i) \\ N_i^* &= K_i - \alpha_{ij} N_j \quad \text{and similarly } N_j^* = K_j - \alpha_{ji} N_i \end{aligned}$$

Note intuitive meaning of α_{ij} !

Note also that we can't solve for one species without knowing the other.

$\Rightarrow$ need to solve for N_i^* and N_j^* jointly.

Plug solution of N_j^* into solution for N_i^* and vice versa.

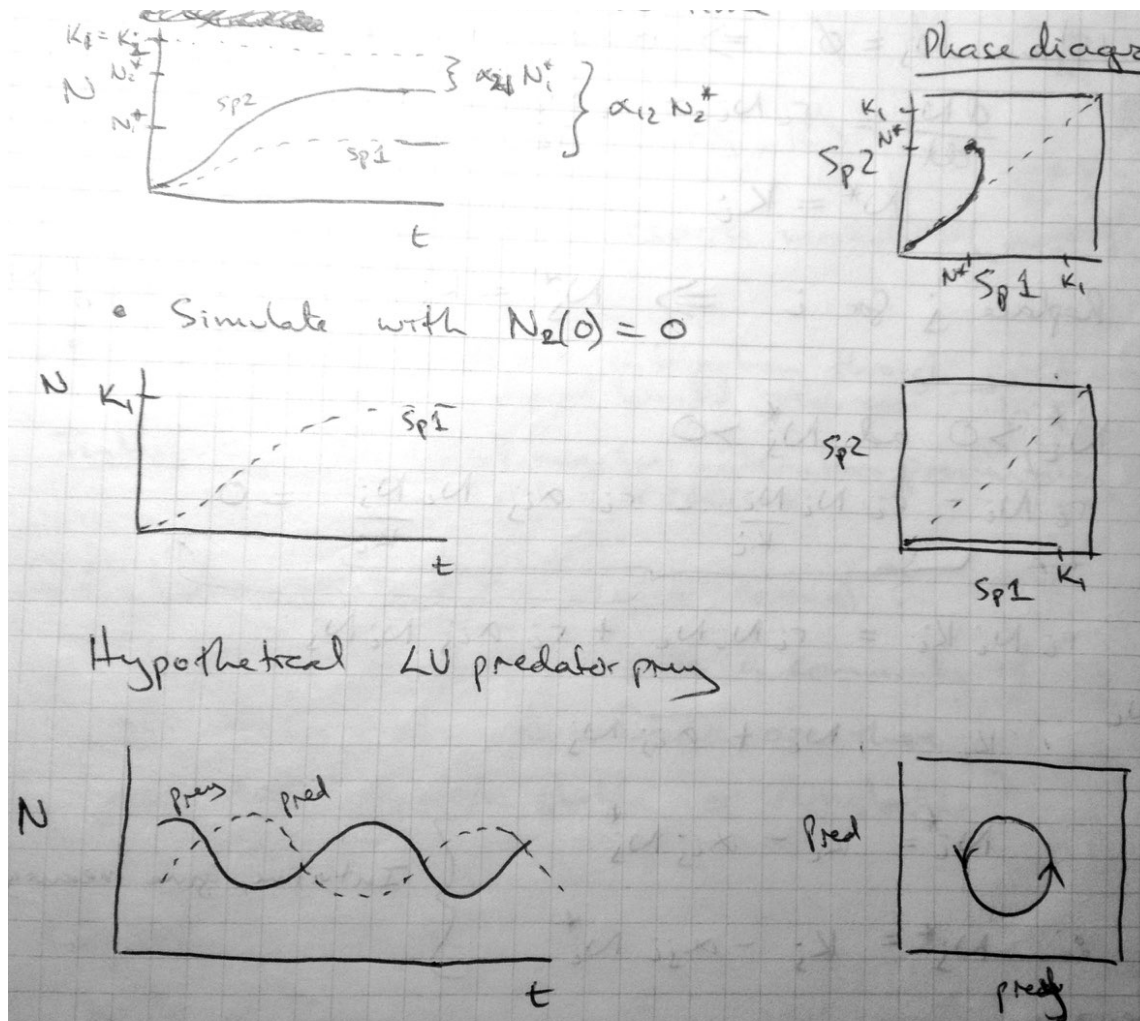
$$\begin{aligned} N_i^* &= K_i - \alpha_{ij} N_j^* \\ &= K_i - \alpha_{ij} (K_j - \alpha_{ji} N_i^*) \\ &= K_i - \alpha_{ij} K_j + \alpha_{ij} \alpha_{ji} N_i^* \\ N_i^* - \alpha_{ij} \alpha_{ji} N_i^* &= K_i - \alpha_{ij} K_j \\ N_i^* (1 - \alpha_{ij} \alpha_{ji}) &= K_i - \alpha_{ij} K_j \\ N_i^* &= \frac{K_i - \alpha_{ij} K_j}{1 - \alpha_{ij} \alpha_{ji}} \end{aligned}$$

Phase portraits/diagrams

R-code demonstration

Work through ode-solver code

Simulate coexistence over time



Graphical Analysis

Earlier we showed that from $\frac{dN_i}{dt} = 0$ that:

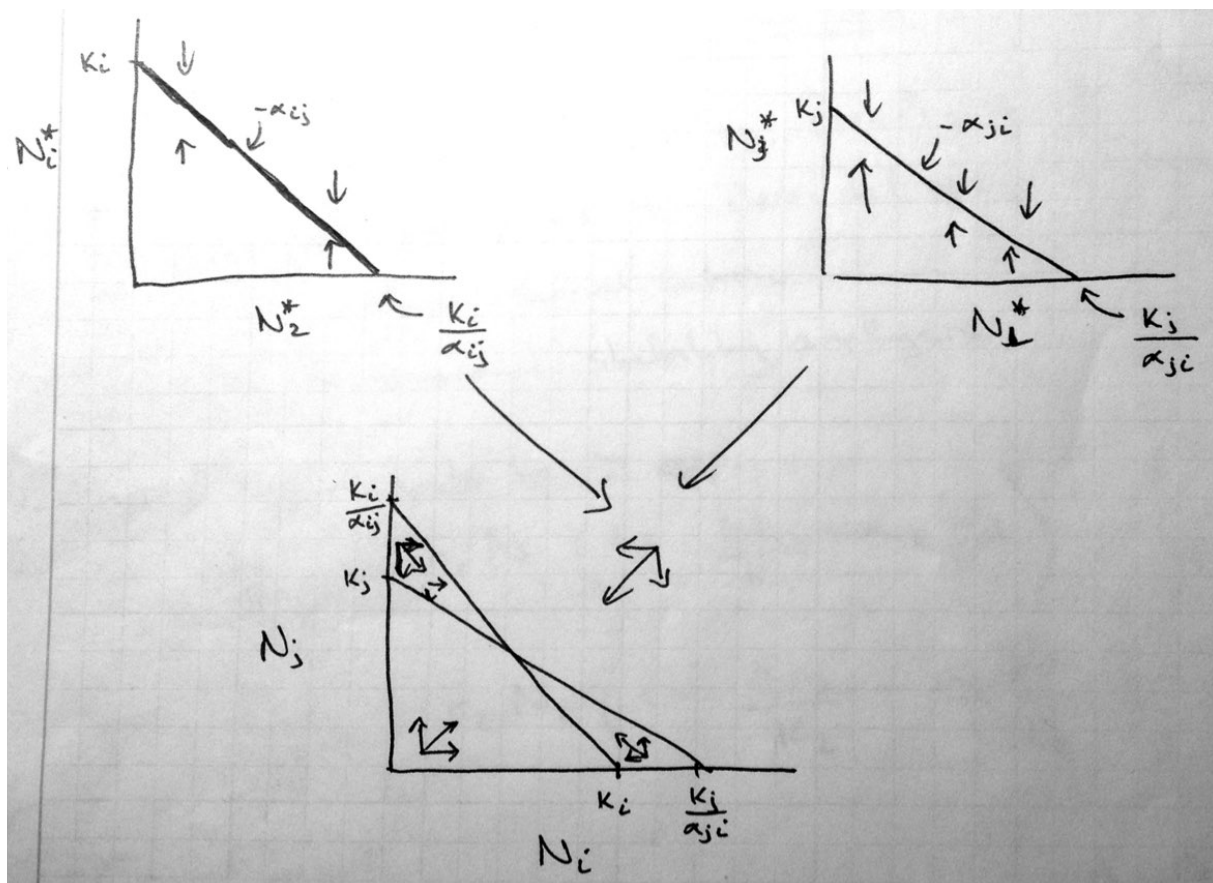
$$N_i^* = K_i - \alpha_{ij} N_j$$

$$\Rightarrow N_j = \frac{K_i - N_i^*}{\alpha_{ij}}$$

Thus isocline intersects x-axis at

$$N_j = \frac{K_i - 0}{\alpha_{ij}} = \frac{K_i}{\alpha_{ij}}$$

Same goes for 2nd species.



Inferences/Conclusions:

- When can both spp. coexist?
- When can sp 1 out-compete sp 2?
- When can sp 1 invade sp 2?

Coexistence: intra > inter for both species
 (i.e. $\frac{K_i}{\alpha_{ii}} = \frac{K_i}{1} < \frac{K_j}{\alpha_{ji}} \Rightarrow \alpha_{ii} > \alpha_{ji}$)

Sp i dominance: intra > inter for i & intra < inter for j

Priority effect: intra < inter for both
 (i.e. $\frac{K_i}{\alpha_{ii}} = \frac{K_i}{1} > \frac{K_j}{\alpha_{ji}} \Rightarrow \alpha_{ii} < \alpha_{ji}$)

Simulations

R-demonstration - Walk through first set of parameter values.

With $r_1 = r_2 = K_1 = K_2 = 1$ and $N_1(0) = N_2(0) = 0.01$

α_{21}	α_{12}	Outcome	Initial
0.5	0.7	coexist	
1.5	0.5	sp 1	
0.5	1.5	sp 2	$N_1(0) = N_2(0)$
1.5	1.7	sp 2	
2	2	sp i	$N_i(0) > N_j(0)$

$\Rightarrow$ Priority effect
 $\Rightarrow$ Alternative stable states

R-code: Work through other parameter values

Show `VectorField.R`